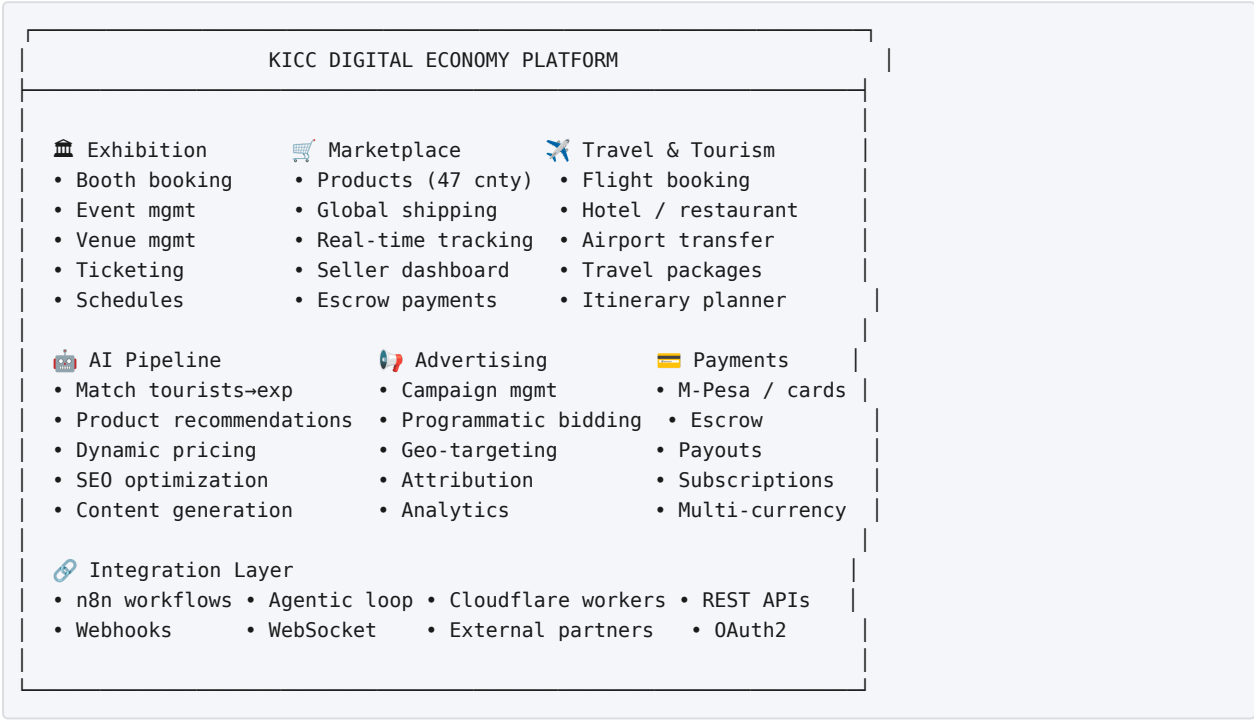


KICC Digital Economy Platform — Architecture Blueprint

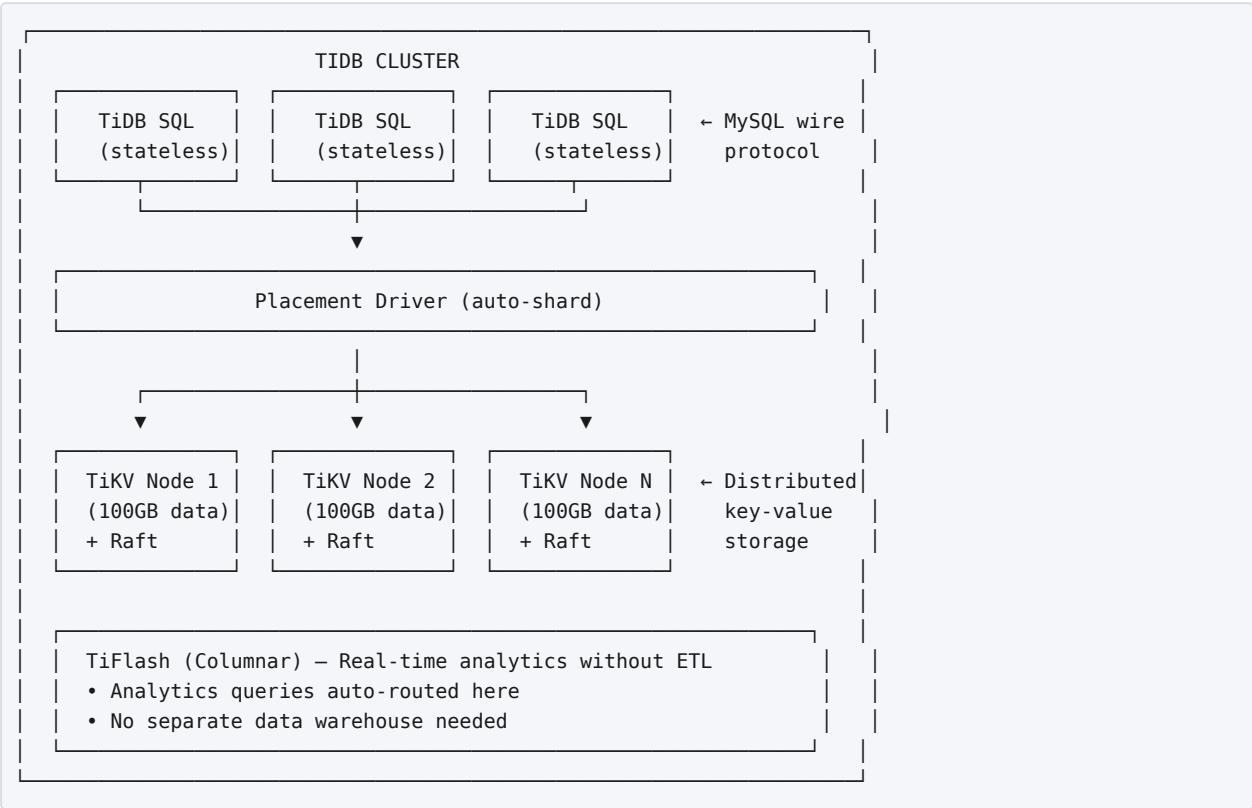
Platform Vision

A unified **Kenyan Digital Economy Super-App** connecting:



Architecture Layers

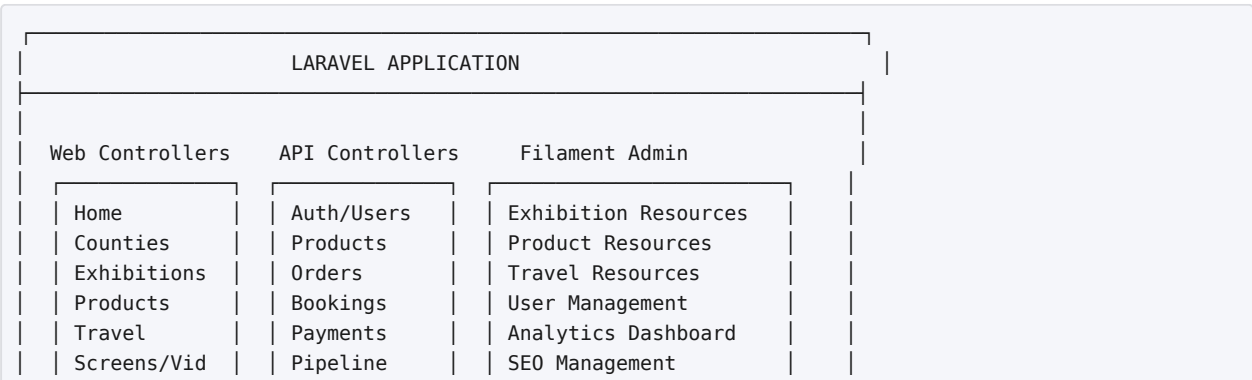
Layer 1: Database (TiDB — MySQL Compatible Distributed)

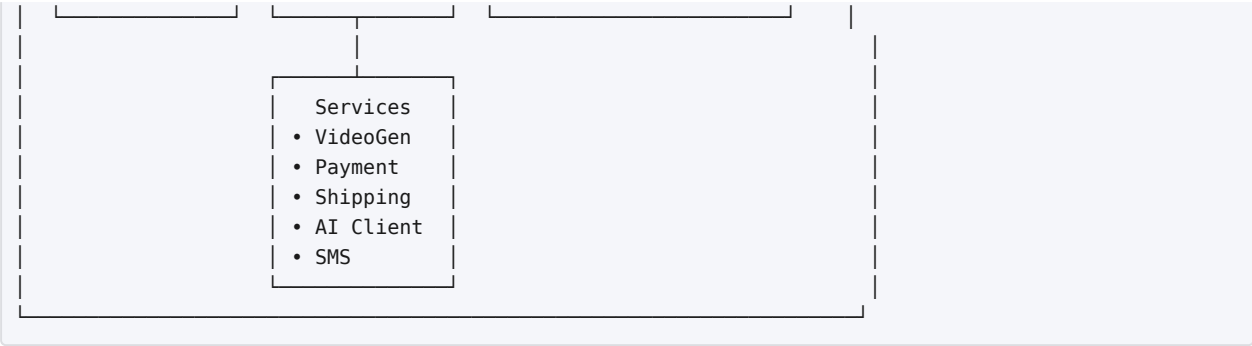


Why TiDB over Aurora MySQL:

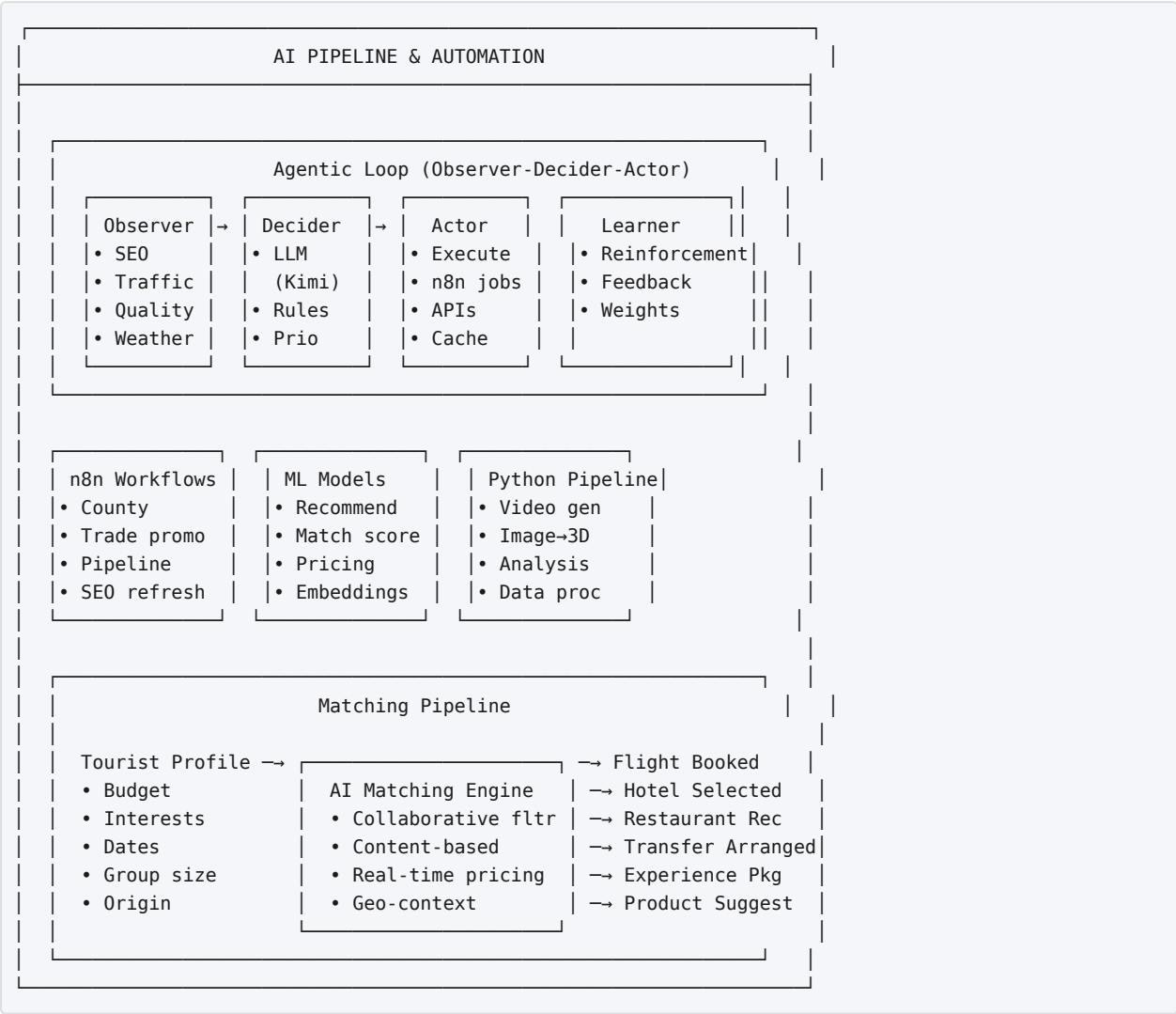
Capability	Aurora MySQL	TiDB
Max storage	256 TiB (single node write bottleneck)	Petabyte+ (auto-shards across nodes)
Write scaling	Single primary node	Multi-node writes — linear scaling
Multi-region	Read replicas only	Active-active across regions
HTAP	Separate Redshift/Snowflake needed	Built-in columnar (TiFlash) — analytics on live data
MySQL compat	✅ Full	✅ Wire-protocol compatible
Zero-downtime scaling	Vertical (resize instance)	Horizontal (add nodes, no downtime)
Cost at petabyte scale	Very expensive (large instances)	Commodity hardware — scales linearly

Layer 2: Application (Laravel)

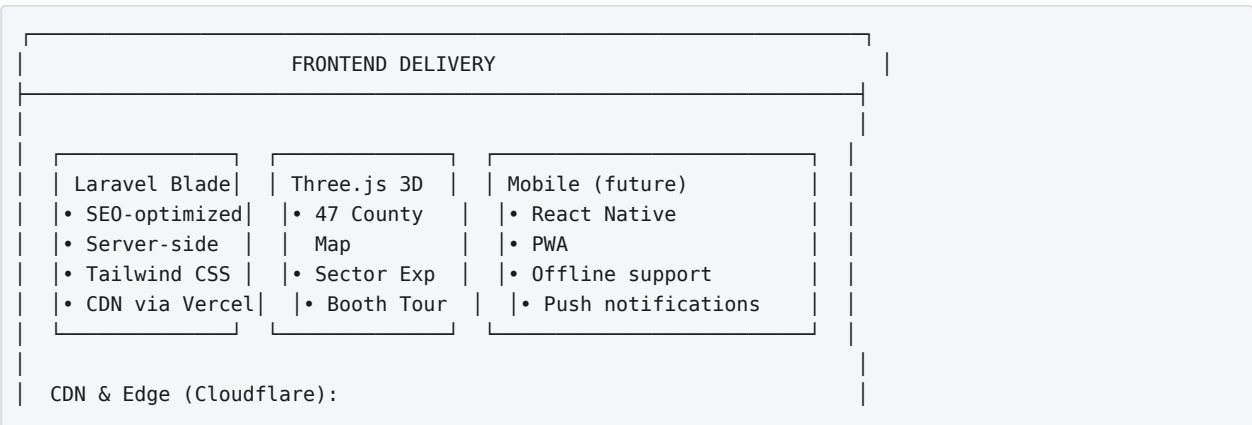




Layer 3: AI Pipeline & Automation



Layer 4: Frontend Delivery



- Static assets • 3D HTML • Videos • Images • API caching
- DDoS protection • SSL • Workers for geo-routing

Domain Model — 20 Business Domains (~200 tables)

Core Platform (56 existing tables — kept as-is)

E-Commerce Marketplace (new)

Table	Purpose
products	All Kenyan products across 47 counties
product_categories	Category tree (agri, crafts, textiles, etc.)
product_variants	Size, color, weight variants
product_images	Gallery photos
product_reviews	Buyer ratings
suppliers	County-based suppliers
supplier_products	Supplier→product mapping
inventory	Stock per warehouse/variant
warehouses	Fulfillment centers
shopping_carts	Active carts (per user)
cart_items	Items in cart
orders	Customer orders
order_items	Line items
order_status_history	State machine log
shipments	Fulfillment records
shipment_tracking	Courier tracking events
return_requests	RMA process
wishlists	Saved items
coupon_codes	Discount campaigns
price_history	Pricing analytics

Travel & Tourism (new)

Table	Purpose
airlines	Partner airlines
airports	All Kenyan airports
flights	Flight schedules/routes
flight_inventory	Seat availability
flight_bookings	Booking records
flight_passengers	Passenger manifests
hotels	Partner hotels
hotel_rooms	Room types/inventory
hotel_bookings	Booking records
hotel_room_bookings	Room→booking mapping
restaurants	Partner restaurants
restaurant_tables	Table layouts
restaurant_menu_items	Menu catalog
restaurant_bookings	Table reservations
airport_transfers	Transfer services
transfer_bookings	Pickup/dropoff records
travel_packages	Bundled experiences
travel_itineraries	Day-by-day plans
attractions	Tourist sites
attraction_tickets	Entry tickets

AI Pipeline (new)

Table	Purpose
pipeline_jobs	Job definitions
pipeline_runs	Execution history
pipeline_logs	Step-by-step logs
ml_models	Model registry
model_versions	Version tracking
embeddings	Vector embeddings (product, user, content)
recommendations	Generated recommendations
recommendation_logs	Impression/click/conv logging
user_segments	ML-generated segments
match_scores	User→item match scores
a_b_tests	Experiment tracking
feature_flags	Rollout management

Advertising (new)

Table	Purpose
ad_campaigns	Campaign definitions
ad_groups	Ad group hierarchy
ad_creatives	Image/video/text ads
ad_targeting	Audience targeting rules
ad_placements	Slot definitions
ad_impressions	Impression log (high volume)
ad_clicks	Click log
ad_conversions	Conversion tracking
ad_budgets	Budget pacing
ad_publisher_payouts	Revenue share

SEO & Content (new)

Table	Purpose
seo_metadata	Per-page SEO metadata
content_pages	CMS pages
content_blocks	Reusable content
redirects	URL redirect rules
sitemap_urls	Sitemap tracking
search_queries	Internal search log
keyword_tracking	Search ranking history

Logistics & Shipping (new)

Table	Purpose
courier_partners	Shipping carriers
shipping_zones	Regional zones
shipping_rates	Rate cards
shipping_labels	Generated labels
pickup_requests	Supplier pickup
delivery_attempts	Failed delivery log

Payments (expanded)

Table	Purpose
payment_gateways	Gateway configs
payouts	Seller/partner payouts
settlements	Batch settlement records
refunds	Refund processing
disputes	Payment disputes
transaction_logs	Raw gateway logs

Subscriptions (expanded)

Table	Purpose
subscription_features	Feature catalog
subscription_plan_features	Plan→feature mapping
usage_logs	API/feature usage
billing_cycles	Invoice cycles
invoices	Generated invoices
invoice_items	Line items

Database Recommendation: TiDB

TiDB is the optimal choice for this platform because:

1. **MySQL wire-protocol compatible** — Your existing Laravel + 56 tables work with **zero code changes**. Same Eloquent queries, same migrations, same everything.
2. **Auto-sharding (no sharding logic)** — TiDB automatically distributes data across nodes. When you grow from 10 GB to 10 TB to 1 PB, you just add nodes. No manual partitioning, no Vitess complexity.
3. **Petabyte-scale HTAP** — TiFlash columnar replicas let you run real-time analytics directly on production data. No ETL to Snowflake/Redshift. Analytics queries auto-route to TiFlash.
4. **Multi-region active-active** — Deploy in US-East, EU-West, Asia-Pacific simultaneously. Users get local read/write with automatic conflict resolution.
5. **Linear write scaling** — Unlike Aurora's single-primary bottleneck, TiDB writes scale linearly with nodes. 100M transactions/day? Add nodes.
6. **Zero-downtime operations** — Add indexes, change schema, scale up/down without locking. Critical for a 24/7 marketplace.
7. **Cost at scale** — Commodity hardware. 10 nodes of 8 CPU/32GB RAM can handle 1 TB+ for ~\$500/month on bare metal, much less than Aurora at that scale.

Migration Path

SQLite (local dev) → TiDB Serverless (free tier, 5GB) → TiDB Dedicated (prod)

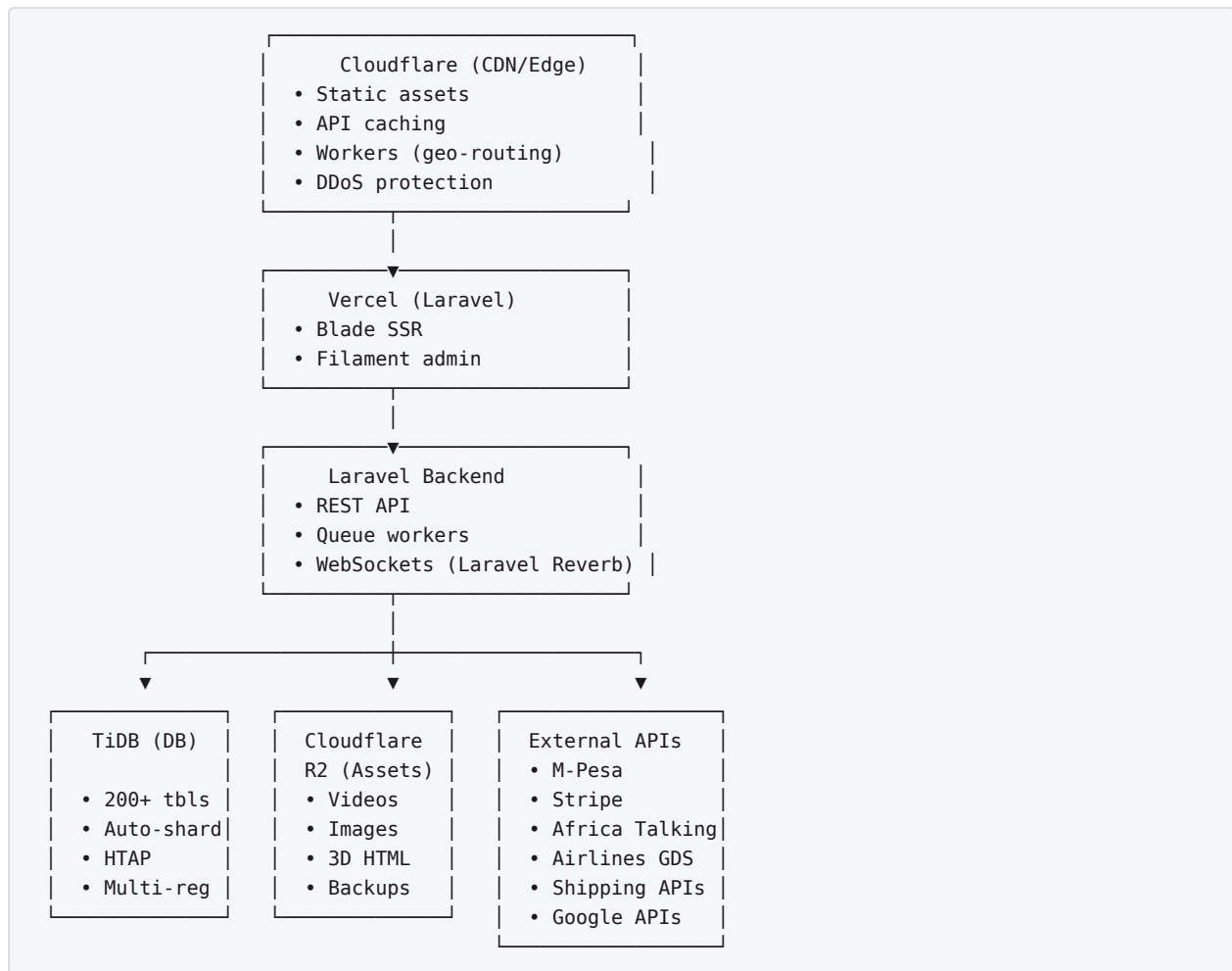
1. **Local:** SQLite (already configured) — for development
2. **Staging:** TiDB Serverless (free up to 5GB, no server to manage)
3. **Production:** TiDB Dedicated cluster (auto-scale, multi-region)

Migration is a single `.env` change:

```
DB_CONNECTION=mysql
DB_HOST=gateway01.us-west-2.prod.aws.tidbcloud.com
DB_PORT=4000
DB_DATABASE=kicc
DB_USERNAME=xxxxx.root
DB_PASSWORD=xxxxx
```

Zero code changes. Same Laravel migrations. Same Eloquent.

Integration Architecture



Data Flow Examples

Tourist Books Kenyan Experience

1. User browses counties/tourism (Blade SSR via Vercel)
2. Agentic loop observes -> triggers recommendation refresh
3. AI pipeline scores user against: flights, hotels, restaurants, products
4. User sees personalized recommendations
5. Books: flight (GDS API) + hotel (TiDB) + restaurant (TiDB table booking)
6. Pipeline auto-books airport transfer (matching driver)
7. Payment: M-Pesa/Stripe -> TiDB transaction log
8. Order: Kenyan product recommended -> added to cart -> shipped
9. Agentic loop learns from booking -> updates match weights
10. Notification: SMS (Africa Talking) + email

Seller Lists Product

1. Seller from e.g., Kiambu lists avocado oil
 2. Product written to TiDB (auto-sharded by product_id)
 3. n8n workflow triggers: SEO metadata generation (agentic loop)
 4. AI pipeline generates embedding vector -> stored in TiDB
 5. Product visible in marketplace with global shipping calculation
 6. TiFlash columnar replica enables real-time seller analytics
 7. When ordered: inventory decremented, shipment created
 8. Courier partner API called for pickup
 9. Tracking events flow back through webhooks
-

Deployment Strategy

Phase 1 (Now)	Phase 2 (Next)	Phase 3 (Scale)
• SQLite local • Laravel serve • Basic features • 56 tables • 3D experiences	• TiDB Serverless • Vercel deployment • E-commerce launch • Travel booking • AI pipeline v1 • n8n automation • M-Pesa integration	• TiDB Dedicated cluster • Multi-region deployment • Global CDN + edge • 200+ tables • Full ML pipeline • Petabyte-scale • 24/7 SRE

Project Structure

```

/home/kicc/Desktop/kicc/
├── kicc-platform/           ← NEW: Full self-hosted Laravel project
│   ├── app/                PHP backend (32 models + 20 controllers + services)
│   ├── database/
│   │   ├── schema.sql      56 current tables
│   │   ├── schema-commerce.sql E-commerce + travel + ads + pipeline tables
│   │   └── schema-full.sql  Complete ~200 table schema
│   ├── public/3d/          Three.js experiences
│   └── setup.sh             One-command setup
├── kenya-3d-platform/      ← EXISTING: Vercel deployment (unchanged)
│   ├── laravel-backend/    Original Laravel for Vercel
│   ├── pipeline/           Python pipeline (video gen, image analysis)
│   ├── agentic_loop/       AI agent system
│   ├── n8n-workflows/      Automation workflows
│   ├── seo-engine/         SEO optimization
│   ├── cloudflare-worker/  Edge functions
│   └── scripts/            Utility scripts
└── start-preview.sh        Static preview server (port 8081)

```